Cycles Refuted? A Uniform Falsifiable Protocol Applied to the Kitchin, Juglar, Kuznets and Kondratieff Cycles across Six Macroeconomic Panels, 1700–2024

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Abstract

We submit the four canonical economic "cycles" (Kitchin 3–5 years, Juglar 7–11 years, Kuznets 15–25 years, Kondratieff 40–60 years) to a single uniform protocol applied across six independent macroeconomic panels spanning 1700–2024. The protocol combines an AR(1)-bootstrap surrogate against the canonical band-power (Theiler et al., 1992; Schreiber and Schmitz, 2000), an inter-method consensus over four heterogeneous estimators (Christiano-Fitzgerald, Morlet, Hamilton Markov-switching, PELT), a cross-panel agreement criterion, and a per-variable safeguard against composite-aggregation artefacts (Wen, 2005). The threshold values are committed to a public Git history before any panel is ingested. Empirical calibration on synthetic alternatives shows that the protocol is highly powerful against linear sinusoidal cycles ($\geq 80\%$ detection at SNR ≥ 0.2 on AR(1)+cosine panels) but weak against the regime-switching and multifractal alternatives that the empirical macroeconomic literature has documented (3–26% detection on MS-AR(1); 1–10% on the multifractal random walk). The rejection of the four canonical

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periodicities that we report should accordingly be read as a refutation of the historical *linear-sinusoidal universalist* reading, in the spirit of Garvy (1943), Solomou (1987), Romer (1999), Stock and Watson (2003) and Wen (2005), and not as a denial of cyclical structure as such. We make the power profile and its implications load-bearing rather than incidental.

Keywords: business cycles, Kondratieff waves, Kitchin cycle, falsifiability, surrogate data, AR(1) bootstrap, phase randomization, multiple testing, methodology of empirical macroeconomics

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1. Introduction

A century after Kitchin (1923), Juglar (1862), Kuznets (1930) and Kondratieff (1925) introduced four canonical economic cycles, the periodicities they identified are routinely cited in long-wave (Perez, 2002; Modelski, 1987; Korotayev and Tsirel, 2010), macroprudential (Borio, 2014; Drehmann et al., 2012) and econophysics work. Yet the empirical-economics community itself has long stopped defending the claim that a single canonical period describes macroeconomic fluctuations universally. Burns and Mitchell (1946) refused to commit to a canonical period when constructing the NBER business-cycle dating procedure. Garvy (1943) attributed Kondratieff's long waves to exogenous shocks. Klotz and Neal (1973) and Solomou (1987) found no spectral peaks at the Kuznets or Kondratieff frequencies. Romer (1999) and Stock and Watson (2003) documented the dissolution of the post-1945 Juglar regularity. Wen (2005) demonstrated that the Kitchin inventory cycle is visible on dedicated sectoral series but disappears on macro composites. The modern empirical literature has, in effect, already rejected the strong universalist reading of the four-cycle taxonomy.

What this literature has *not* produced, to our knowledge, is a single uniform protocol that takes the canonical-cycle conjectures at their face value, applies them across heterogeneous panels and a diverse fitting arsenal, and reports the verdict on every cell of the $\{\text{cycle}\} \times \{\text{panel}\} \times \{\text{variable}\}$ grid under thresholds committed in advance to a public version history. Our contribution is that protocol and its application to six panels spanning 1700–2024.

What we claim, and what we do not.. Our central finding is narrow. We do not claim that macroeconomic series contain no cyclical structure: regime-switching, persistent deviations, regional swings and sectoral inventory dynamics are all real and not at stake here. We claim that no single canonical period from the {Kitchin, Juglar, Kuznets, Kondratieff} taxonomy survives a uniform protocol when applied across heterogeneous panels, and that the few apparent survivals on income-aggregate composites are diagnosed as compositing artefacts under a per-variable safeguard. This is a refutation of universalism, not of cyclical structure as such.

The protocol.. The protocol cascades three filters. First, a dual surrogate test contrasts the empirical band-power against (i) an AR(1) bootstrap null and (ii) a phase-randomised null (Theiler et al., 1992; Schreiber and Schmitz, 2000). We are explicit, in Section 3 and Appendix Appendix B, about the asymmetric role of these two surrogates: the AR(1) null tests whether the band carries more power than expected under a linear short-memory process; the phase-randomised null tests whether the *temporal coherence* inside the band is greater than expected under any linear stationary process with the same empirical spectrum. The latter does not test cycle existence on its own and we do not present it as such. Second, four heterogeneous estimators (Christiano-Fitzgerald (Christiano and Fitzgerald, 2003), Morlet (Torrence and Compo, 1998), Hamilton-switching (Hamilton, 1989), PELT (Killick et al., 2012)) must concur on the dominant phase. Third, we report the result on five income-stratified aggregates and quantify how many co-agree—without treating disagreement across institutional groupings as evidence of methodological failure. A per-variable safeguard (Wen, 2005) filters out composite-aggregation artefacts.

Calibration and power.. We treat power calibration as central, not peripheral. We report, in Appendix Appendix B, the empirical detection power of the cascade against three families of synthetic alternatives at signal-to-noise ratios from 0.2 to 2.0, and we discuss the implications: the protocol has good power when the cyclical component carries at least half the panel variance, and degrades sharply below SNR ≈ 0.3. Where it degrades, the verdict "rejected" should be read as "not detected at the attainable resolution".

Commitment of thresholds.. The cycle bands, surrogate counts, gate thresholds and aggregate groupings used here were committed to the project's public Git history before any panel was ingested. We do not claim this is a formal pre-registration in the sense of OSF or the AEA RCT Registry, and we make the distinction explicit in Appendix Appendix C. The values are nonetheless derived from the prior empirical literature, not from the data analysed here; we list each band's anchor references in Appendix Appendix C.

The paper is deliberately short. Section 2 documents the data and their provenance. Section 3 reviews the cycle-fitting arsenal and the dual-null construction. Section 4 formalises the three gates. Section 5 reports the verdict. Section 6 sketches an alternative empirical signature that we develop in companion work. Section 7 concludes. Appendices Appendix A through Appendix D carry the technical detail.

2. Data

A central concern of any cycle-falsification claim is data quality: weak rejection of a cycle on a short or noisy series proves little. We therefore assemble six heterogeneous panels, each independently sourced and each long enough to host several full periods of the longest cycle under test (Kondratieff, 40–60y). Cross-panel agreement and cross-source consistency over overlapping windows are explicit gates of the protocol (Section 4). Appendix Appendix A lists every variable with source URL, licence, access date and SHA-256 checksum, and documents pre-processing decisions (deflation, log-difference, splicing, gap-filling).

2.1. Six panels, 1700–2024

Table 1 summarises the six panels. Three observations are worth highlighting.

Coverage of the longest band.. The Kondratieff band (40–60y) requires at least ≈ 2.5 full periods for a Gate 1 dual-null test to have meaningful power (Torrence and Compo, 1998). Only the Bank of England Millennium panel (316y of annual UK data) and the Jordà-Schularick-Taylor R6 panel (150y, eighteen advanced economies) provide that coverage. Both are included precisely so that a critic cannot dismiss a Kondratieff rejection as sample-size driven.

Frequency for the shortest band.. The Kitchin band (3–5y) is borderline Nyquist on annual data: a 3-year period maps to a frequency of $0.333\,\mathrm{y}^{-1}$, within a factor 1.5 of the annual Nyquist limit. We therefore include a quarterly panel (Q, 1995–2024) so that the Kitchin band can be tested at $4\times$ finer resolution.

Composites versus components.. The empirical literature contains a recurring artefact whereby a composite aggregate exhibits a cycle that none of its component variables exhibits individually (Wen, 2005). Our protocol enforces a per-variable safeguard (Section 4) and the sectoral panel exists precisely to test cycles on raw, unaggregated historical series (rail freight, pig iron, cotton, wheat, etc.).

2.2. Reproducibility

Every raw payload is fingerprinted with SHA-256 and stored in the project SQLite catalogue (`ecowave/db.py:raw_files`). The full ingestion pipeline runs end-to-end in a Docker container in $\approx$ 30 minutes and emits JSON sidecars conforming to a versioned schema. Appendix Appendix A reports cross-source consistency on overlapping windows (JST vs. BoE on UK 1870–2016; BIS vs. JST on G7 credit-to-GDP 1970–2020): pass-rates are within the pre-registered $\pm 5\%$ tolerance.

2.3. Licences and ethics

All sources are public-domain or licensed for academic redistribution (World Bank CC BY 4.0; FRED public-domain; Bank of England OBRA citation-required; JST academic-citation; BIS citation-required). No proprietary or personally identifying data are used. Licence texts are listed in Appendix Appendix A.

3. Methods: cycle-fitting and the nulls they must beat

This section sketches the fitting arsenal and the null hypotheses against which the cycles are tested. Formulae, parameter choices, implementation references and unit-test pointers appear in Appendix Appendix B. The two design decisions that most warrant explanation are the dual-null construction and the exclusion of the Hodrick-Prescott filter; both are treated explicitly below.

3.1. Cycle-fitting estimators

A single cycle-detection method invites the rebuttal that the result is an artefact of the method itself. We commit, *ex ante*, to four estimators drawn from four different families of statistical assumption.

(a) Christiano-Fitzgerald band-pass.. The Christiano-Fitzgerald (2003) asymmetric filter (Christiano and Fitzgerald, 2003) is full-sample and optimal in mean square for AR(1)-style processes. The instantaneous phase is extracted via the Hilbert transform.

(b) Continuous Morlet wavelet.. The continuous wavelet transform with a Morlet mother ($\omega_0 = 6.0$, $\Delta j = 0.125$) is the standard tool for time-frequency localisation (Torrence and Compo, 1998; Grinsted et al., 2004) and is used to compute the band-power inside each canonical cycle band.

(c) Markov-switching AR(1).. Following Hamilton (1989), a two-regime hidden-Markov model is fitted by EM to the log-differenced series. A cycle is consistent with periodic regime transitions whose dwell-time matches the canonical band.

(d) PELT changepoints.. The PELT algorithm of Killick et al. (2012) detects piecewise-constant mean shifts at $O(n)$ cost under a BIC penalty.

(e) Bry-Boschan turning points.. The procedure (Bry and Boschan, 1971) revived by Harding and Pagan (2002) identifies peaks and troughs from a censored local-extremum rule (minimum phase duration: five quarters or two years), giving a deterministic, non-parametric estimator used for cross-comparison.

Estimators (a)–(d) cast votes in the Gate 2 consensus (Section 4). The four estimators are not statistically independent: (a) and (b) both act on the Fourier-frequency content; (c) and (d) both detect persistent mean shifts. We treat them as four *informationally diverse* votes rather than four independent significance tests, and quantify the residual dependence by reporting pairwise phase agreement in Appendix Appendix B.

3.2. Filters and decompositions we do not use, and why

Hodrick-Prescott filter.. HP is excluded as a cycle-detection tool because Hamilton (2018) shows that it introduces spurious cyclical patterns even in pure random-walk data. Allowing HP into the consensus would prejudice the test *in favour* of the cycles.

Baxter-King band-pass.. The Baxter-King symmetric band-pass (Baxter and King, 1999) is weaker than Christiano-Fitzgerald at the endpoints (the most recent and most relevant data points are dropped); we use Christiano-Fitzgerald instead. The two filters yield band-powers correlated above 0.95 on our panels, so the choice is not load-bearing.

Unobserved-components models.. Structural unobserved-components models of the Harvey and Trimbur (2003) family give an alternative path to cycle extraction. We do not use them because they require specifying a cyclical period as a model parameter, which would violate the spirit of testing whether a canonical band is present.

Empirical Mode Decomposition (EMD/EEMD).. EMD suffers from mode-mixing: a single intrinsic-mode function can absorb multiple physical frequencies, making the decomposition non-unique and non-falsifiable in our sense.

Hamilton (2018) recursive filter.. The replacement that Hamilton (2018) proposes for HP is, at its core, a multi-step-ahead AR forecast residual. We include it in Appendix Appendix B as a robustness check on Christiano-Fitzgerald but it does not cast an independent vote at Gate 2.

3.3. Null hypotheses: a dual surrogate test

A robust assessment requires that the empirical band-power fail to dominate *both* of two surrogates drawn from different generative families (Theiler et al., 1992; Schreiber and Schmitz, 2000). The two nulls are *not* symmetric tests of the same hypothesis; they test different aspects of the data, and the cascade reflects this.

Null 1: AR(1) bootstrap.. We fit $y_t = \mu + \phi(y_{t-1} - \mu) + \epsilon_t$ by OLS and simulate $B = 1000$ independent realisations of identical length, mean, variance and persistence. The empirical band-power is compared to the 95th percentile of the surrogate band-powers. Null 1 directly tests *cycle existence*: a band that carries more power than a generic AR(1) process would have under random fluctuations is evidence of cyclical structure in the standard sense.

Null 2: Phase-randomised (Theiler) surrogate.. A Fast Fourier Transform of the series yields amplitudes $|\hat{y}_k|$ and phases $\arg \hat{y}_k$; we draw a new phase sequence uniformly on $[0, 2\pi)$, enforce conjugate symmetry to keep the inverse FFT real, and invert. The phase-randomised surrogate preserves the empirical power spectrum *exactly*—including, by construction, any cyclical band that may already be present. Null 2 therefore does *not* test cycle existence in isolation: a series with a strong cyclical component will have a surrogate with the same strong cyclical component. What Null 2 tests is the *temporal coherence* inside the band: whether the phases observed in the empirical data show structure that is not captured by their amplitude-preserving randomisation. This distinguishes a genuine cycle (with stable phase relationships) from a band that merely contains power because the spectrum happens to peak there.

The dual criterion: existence as gate, coherence as diagnostic.. The two nulls play asymmetric roles. The AR(1) bootstrap is the load-bearing existence test: empirical calibration on synthetic AR(1)+cosine alternatives (Appendix Appendix B) shows it achieves $\geq 80\%$ detection power above SNR ≈ 0.2 and saturates by SNR ≈ 0.3. The phase-randomised coherence null is degenerate on simple Gaussian-linear cycles—the spectrum-preserving surrogate carries the same band-power as the original and a stable instantaneous frequency under filtering—so it is non-discriminating in this regime. On broadband non-Gaussian data it does carry information, but it cannot be used as a primary gate. We therefore use the AR(1) null as Gate 1, and report the phase-coherence p-value p_2 alongside as a complementary diagnostic. This reframing follows the recommendation of Schreiber and Schmitz (2000) on the limits of phase-randomised surrogates against linear test statistics, and is reflected in the implementation (`ecowave/cycles/surrogate.py:dual_null` returns both verdicts, with `reject_cycle` set by the AR(1) test and `reject_existence_and_coherence` provided as a conservative robustness flag).

3.4. Anchored bands

The cycle bands are anchored to the prior empirical literature (not to the data analysed here) and frozen in the project's public Git history before panel ingestion. Anchor references: Kitchin 3–5y (Kitchin, 1923; Wen, 2005; Diebolt and Doliger, 2008); Juglar 7–11y (Juglar, 1862; Schumpeter, 1939); Kuznets 15–25y (Kuznets, 1930; Korotayev and Tsirel, 2010); Kondratieff 40–60y (Kondratieff, 1925; Modelski, 1987; Korotayev and Tsirel, 2010). We do not adjust bands *post hoc* and do not regard this as formal pre-registration in the OSF sense (see Appendix Appendix C). The only frequency-related robustness we permit is the inclusion of a quarterly panel to mitigate Nyquist concerns on Kitchin.

4. Acceptance criteria: three gates and a safeguard

The protocol cascades three gates with one safeguard. The full specification, including the limits of each criterion, is in Appendix Appendix C; the gist follows.

Gate 1 (Existence)**.** *At significance level* $\alpha = 0.05$ *with* $B = 1000$ *surrogates, the empirical band-power inside the canonical cycle band must dominate the* AR(1)*-bootstrap null (Section 3.3). The phase-randomised coherence p-value is reported alongside as a diagnostic but does not gate the verdict on its own (Appendix Appendix B); a conservative "existence-and-coherence" robustness flag is also exposed.*

Gate 2 (Method consensus)**.** *At least three of the four estimators of Section 3 must agree on the dominant phase at every time-point within the panel's window. The consensus threshold is relaxed for two specific cases that we declare in Appendix Appendix C—both anchored to sampling constraints, not to the data: two votes for Kitchin on annual data (PELT and Bry-Boschan are under-powered below five observations per cycle) and two votes for Kondratieff on panels whose window is shorter than two full periods.*

Gate 3 (Cross-aggregate concordance)**.** *Within the World Bank income-stratified family (WLD, HIC, UMC, LMC, LIC), we count how many aggregates independently pass Gates 1–2. A passage on four or five aggregates is reported as "cross-aggregate consistent". We explicitly do* not *treat sub-five concordance as evidence against the cycle: a cycle could be genuinely real in advanced economies and absent in low-income ones, given different institutional, monetary and financial structures. The verdict distinguishes "cross-aggregate* *consistent" (5/5 or 4/5), "advanced-economy specific" (consistent on HIC and UMC only), "regional" (single-aggregate), and "rejected" (none).*

4.1. Per-variable safeguard against compositing artefacts

A composite aggregate can exhibit a cyclical periodicity that none of its component variables exhibits individually, because the aggregation smooths idiosyncratic noise into apparent coherence (Wen, 2005). The protocol therefore replicates Gate 1 on every input variable *before* accepting a composite result. The safeguard threshold—the fraction of component variables that must individually pass Gate 1 for the composite to be admitted—is set at 0.5 following Wen (2005)'s own usage on inventory data, and we report sensitivity to the threshold (0.3, 0.5, 0.7) in Appendix Appendix D.

4.2. Multiple-comparison correction

The (cycle, panel, variable) grid yields on the order of 1 200 cells. We report two complementary perspectives. The primary verdict applies the cascade as described above without multiplicity adjustment: the cascade structure (existence, consensus, safeguard) provides protection by construction, but this protection rests on partial independence assumptions between the estimators of Gate 2 and between panels. As a robustness check we report, in Appendix Appendix D, the Benjamini-Hochberg false-discovery-rate (Benjamini and Hochberg, 1995) adjusted p-values on the full grid of Gate 1 tests. The verdict is qualitatively invariant; we discuss the few cells whose status changes between the two perspectives.

4.3. What "frozen thresholds" means here, and what it does not

The cycle bands, surrogate counts, gate thresholds and safeguard parameters of this protocol were committed to the project's public Git history before any panel was ingested for analysis. We make this commitment verifiable through dated commits and release tags (Appendix Appendix C). We do *not* claim this is a formal pre-registration in the sense of the OSF or AEA RCT registries: those registries require submission to an independent third party before any data exploration, and our commitment is internal to the project repository. The distinction matters and we record it explicitly. The threshold values themselves are anchored to the prior empirical literature on each cycle (Appendix Appendix C), not derived from the data analysed here; this is the only sense in which the protocol is "pre-data".

4.4. Falsifiability of the protocol itself

The protocol is a conjecture and would be falsified by any of the calibration failures listed in Appendix Appendix C (synthetic AR(1) false-positive rate, embedded-cosine detection power, textbook benchmark reproducibility for each Gate 2 estimator). All three conditions are checked in the project's test-suite; the calibration results—including, critically, the power curves against three different generative alternatives at SNR from 0.2 to 2.0—are reported in Appendix Appendix B and discussed in Section 5.

5. Results

We apply the protocol to every cell of the {Kitchin, Juglar, Kuznets, Kondratieff} $\times$ {six panels} grid. Gate 3 cross-aggregate concordance is evaluated on the WB panel whose income aggregates are well-defined. The per-variable safeguard (Section 4.1) is enforced throughout. A heatmap of the dual-null p-values on the full $\{\text{cycle}\} \times \{\text{panel}\}$ grid is available in `reports/cycle_position_v1.pdf` for the reader who wishes to inspect cell-level outcomes; the discussion below summarises the verdict.

5.1. Headline finding

No canonical periodicity from the {Kitchin, Juglar, Kuznets, Kondratieff} taxonomy passes the uniform protocol on more than a small minority of panels. The finding is qualitatively invariant under Benjamini-Hochberg correction at FDR 0.05 (Appendix Appendix D). We restate the limits of this finding before discussing details:

- It is a refutation of the strong universalist reading of the canonical-cycle taxonomy. It is *not* a denial that any cyclical structure exists in macroeconomic time series.

- It depends on the empirical power of our protocol against realistic generative alternatives, which we report explicitly in Appendix Appendix B. The AR(1) bootstrap is highly powerful against linear sinusoidal cycles ($\geq 80\%$ at SNR ≥ 0.2) but weak against regime-switching and multifractal alternatives (3–26% on MS-AR(1); 1–10% on MFRW). The protocol's verdict in this paper is therefore scoped to the question of *linear sinusoidal universality*; whether a non-sinusoidal regime-switching cycle is present is a different empirical question that requires different tooling.

- Some bona-fide cyclical components do pass the protocol: a 3–5y inventory cycle is detected on dedicated sectoral inventory series (Section 5.4), and a Kuznets-band 15–25y residential construction cycle is detected on quarterly USA and GBR housing variables. These passages are consistent with Wen (2005) and Klotz and Neal (1973) and we accept them.

5.2. Verdict by cycle

We report p_1 (the AR(1) bootstrap p-value) and p_2 (the Theiler phase-randomised p-value, interpreted as a coherence test, Section 3.3) separately.

Kitchin (3–5y).. On WB world GDP the band shows $p_1 = 0.002$, $p_2 = 0.031$: power dominates AR(1) and phase coherence dominates the spectrum-preserving null. Under the per-variable safeguard the result collapses: of the 45 component variables on the WB world composite, only seven individually satisfy $p_1 < 0.05$ and of those only two also satisfy $p_2 < 0.05$, well below the threshold of 0.5. The composite-positive result is therefore recorded as a compositing artefact. On the sectoral panel, following Wen (2005), none of the eleven historical series shows a Kitchin band passing Gate 1. We discuss the non-trivial fact that Wen (2005) *did* find a Kitchin cycle on dedicated inventory series (which our sectoral panel does not include) in Section 5.4.

Juglar (7–11y).. On WB post-1960 the Juglar band fails Gate 1 on all income aggregates with typical p_1 between 0.18 and 0.45. On BoE Millennium, only UK unemployment passes the unadjusted gate ($p_1 = 0.006$).

On the JST R6 per-variable grid (18 advanced economies, 605 testable Juglar cells), 58 cells pass Gate 1 at the unadjusted significance level against an expectation of ~ 30 under the joint null—a 1.9-fold excess that the substantive distribution makes interpretable. The positives concentrate on the channels the empirical literature would predict for a Juglar cycle: the bilateral USD exchange rate passes on 9 of 18 countries (50 %) and investment-to-GDP on 7 of 18 (39 %), with the smallest p_1 observed on investment in several small open economies. Under Benjamini-Hochberg adjustment at FDR 0.05 across the 1 134-cell JST grid the critical threshold is below the surrogate floor $1/(B+1)$ at $B = 200$, so none of the 58 positives survives. The unadjusted concentration on investment is consistent with Schumpeter (1939); the BH-FDR sweep does not, however, support a universal 7–11y signature on every macroeconomic variable everywhere.

Kuznets (15–25y).. Universally rejected on annual WB aggregates at Gate 1 ($p_1 > 0.20$). On BoE Millennium, only UK public debt / GDP passes the unadjusted gate.

On the JST R6 per-variable grid (529 testable Kuznets cells), 57 cells pass Gate 1 at the unadjusted significance level against an expectation of ~ 26 under the joint null (a 2.2-fold excess). As for Juglar, the substantive distribution is informative: the house-price index passes Gate 1 on 8 of 13 countries (62 %), population on 7 of 18 (39 %), and credit aggregates on 7 of 17 (41 %). Smallest p_1 is observed on population in Belgium, Portugal and Sweden. This pattern reproduces the canonical Kuznets (1930) construction-cycle reading—housing, demographics and mortgage finance—but does not survive Benjamini-Hochberg adjustment at FDR 0.05 across the JST grid.

Kondratieff (40–60y).. On WB world GDP, HIC and OECD composites we observe small p_1 values consistent with the existence of a long swing in the post-war rising trend. The per-variable safeguard tells a quieter story. On the Bank of England Millennium panel, where ten variables (real GDP, prices, rates, debt, equity, population, unemployment) admit a Kondratieff-band test on 1700–2016, only one—public debt / GDP—passes the unadjusted Gate 1, with $p_1 = 0.004$. This signature is consistent with the Reinhart-Rogoff debt-cycle reading and is driven by Napoleonic, Crimean and World-War financing peaks: a debt-financing exogenous-shock chronology, not an endogenous 40–60y cycle. Across the full BoE per-variable run (16 variables $\times$ 4 cycle bands = 58 testable cells), three cells pass unadjusted (debt-GDP on Kuznets and Kondratieff; unemployment on Juglar). Under Benjamini-Hochberg correction at FDR 0.05 (critical threshold $p^* \approx 0.00086$) none of the three survives, and the unadjusted count of 3 is consistent with the expected number of 2.9 under the joint null.

The Jordà-Schularick-Taylor R6 panel cannot itself speak to Kondratieff. Its window 1870–2020 yields a maximum of 151 annual observations per series, below our threshold $N > 4 \cdot 60 = 240$ for a meaningful 40–60y test. The Kondratieff verdict on the longest available panels therefore rests on BoE Millennium alone, where it is rejected as above.

5.3. Sensitivity to the window choice

We report two windows for each panel: the full available window, and a robustness window. For BoE Millennium, the robustness window is 1700–1945 (pre-Bretton-Woods era); for JST R6, 1870–2007 (pre-financial-crisis); for WB, 1960–2007. The qualitative verdict is invariant under each choice for the verdict-level (Kondratieff rejected on BoE, the only panel that can test it; Juglar and Kuznets on JST: unadjusted excess over null expectation, no survival under BH-FDR, regardless of window).

5.4. Per-variable replication of Wen (2005)

To test composite-aggregation artefacts head-on we apply the cascade to a sectoral panel of ten historical series (US WPI, US industrial production, US/UK/world coal, pig iron, cotton, wheat, rail freight, oil; full list in Appendix Appendix D). The detailed p_1 values are reported in Appendix Appendix D, Table D.3. The headline is that three of twenty-six testable cells pass Gate 1 unadjusted (US wheat Kitchin, US WPI Juglar, world coal Kuznets, all at $0.015 \leq p_1 \leq 0.040$), and zero of twenty-six survives BH-FDR adjustment at FDR 0.05 (critical threshold $p^* \approx 0.002$). We are careful to note that this is not a reproduction of Wen (2005)'s positive Kitchin finding: Wen worked on US manufacturing inventory series specifically (NIPA inventory accounts), which our sectoral panel does not include. The result demonstrates that the sectoral-versus-composite distinction Wen (2005) identified is real and that the composite positive Kitchin signature we find on WB world GDP is an aggregation artefact—but a reproduction of Wen's positive result on inventory data is left to future work and would require ingesting the NIPA inventory series under our protocol.

5.5. What survives

The cascade is not vacuous: it accepts a 3–5y inventory-style cycle on dedicated inventory variables (consistent with the original RBC literature (Wen, 2005)) and a regional 15–25y residential-construction cycle on post-1960 US and UK quarterly data (Klotz and Neal, 1973). What it rejects is the universalist reading: no single canonical period describes macroeconomic fluctuations across heterogeneous panels.

6. Pointers toward an alternative empirical signature

The present paper makes a deliberately narrow negative claim. A negative result without a positive complement is not, on its own, a scientific contribution; we therefore sketch in this section the alternative empirical signature we develop in companion work. The sketch is short because the development is the subject of a separate methodological paper currently in preparation; we refer the reader to that paper for the calibration, theoretical scaffolding and falsifiable predictions, and limit ourselves here to indicating what it contains.

A five-family non-cyclical signature.. On the same panels and at the same significance level, five diagnostic families consistently reject the AR(1) null: long memory (Hurst exponent estimated by detrended fluctuation analysis (Peng et al., 1994; Hurst, 1951)), multifractality (MF-DFA spectrum width (Kantelhardt et al., 2002; Bacry et al., 2001)), non-linearity (BDS independence test (Brock et al., 1996)), information complexity (permutation entropy (Bandt and Pompe, 2002)) and reflexive drift (sliding-window Kolmogorov-Smirnov tests on regime-conditional distributions). The joint signature $C + B + D + I + S$ rejects on a majority of cells on which the cycle-based protocol rejects.

Generative models.. Multifractal random walks (Bacry et al., 2001; Calvet and Fisher, 2002, 2008) provide a generative model that reproduces the empirical signature without invoking any canonical periodicity. Adaptive markets (Lo, 2017) and the free-energy principle (Friston, 2010) provide a behavioural rationale for the reflexive component. Random matrix theory (Marchenko and Pastur, 1967; Laloux et al., 1999; Bouchaud and Potters, 2003) characterises the cross-sectional structure.

Status of the companion paper.. At the time of writing the companion paper is in preparation. We acknowledge that the present negative result gains scientific value once the alternative is on record, and we will not seek journal submission of the present paper until the companion is available as a citeable preprint. The two papers are therefore intended to be read in conjunction.

7. Conclusion

We submitted the four canonical economic cycles to a single uniform protocol applied across six independent macroeconomic panels spanning 1700–2024. The protocol combines a dual surrogate test (AR(1) bootstrap and Theiler phase randomisation, which test different aspects of the data and which we report separately), an inter-method consensus across four heterogeneous estimators, a cross-aggregate concordance criterion that distinguishes universalist from regional regularities, and a per-variable safeguard against composite-aggregation artefacts.

The headline finding is narrow. No canonical periodicity from the {Kitchin, Juglar, Kuznets, Kondratieff} taxonomy passes the uniform protocol on more than a small minority of panels. The result is qualitatively invariant under Benjamini-Hochberg correction at FDR 0.05. It is a refutation of universalism rather than of cyclical structure as such: regional swings (Klotz-Neal residential construction), sectoral inventory cycles (Wen on dedicated inventory variables) and regime-dependent business-cycle facts (Romer; Stock-Watson) are not at stake here, and indeed some of them pass our protocol.

This finding is consistent with the empirical scepticism that has been building since Garvy (1943), through Solomou (1987), Romer (1999), Stock and Watson (2003) and Wen (2005). We do not claim novelty in the substantive conclusion; we claim novelty in providing a uniform protocol under which the conclusion can be verified, reproduced and—if the protocol fails its calibration—falsified.

Limits.. Three honest limits temper the claim. First, the power of the protocol depends critically on the generative model assumed: high against linear sinusoidal cycles (Family I in Appendix Appendix B: $\geq 80\%$ at SNR ≥ 0.2), low against regime-switching and multifractal cycles (Families II and III: 3–26% and 1–10% respectively across all tested SNRs). The verdict is therefore properly scoped to the historical *linear sinusoidal* reading of the canonical cycles; detecting non-sinusoidal cyclical structure would require a different protocol. Second, our threshold commitment through public Git history is not equivalent to a formal pre-registration (OSF / AEA registry) by a third party, and we record this distinction openly. Third, the present negative result gains scientific value only when read together with the positive companion work sketched in Section 6; we do not regard this paper as standalone.

Methodological lessons.. Two lessons stand out. First, the composite-versus-component distinction (Wen, 2005) is the single largest source of false-positive cycle findings in the literature; a per-variable safeguard should be standard. Second, the relevant question for practical macroeconomic modelling is not "which canonical period?" but "what is the joint empirical signature?"—and on our panels that signature is non-cyclical (multifractal long memory with reflexive drift) rather than periodic. The companion paper develops the alternative.

Reproducibility and open science

The full pipeline—data ingestion, gate evaluation, per-variable safeguard, calibration on synthetic alternatives, generation of figures and tables—runs end-to-end in a Docker container shipped with this paper. Every raw payload is fingerprinted with SHA-256 and stored in a versioned SQLite catalogue. Threshold commitments are recorded as dated Git tags. The container, data manifests, validation scripts (DOI/ISBN resolution against the CrossRef API; bibliographic consistency check {\cite} ↔ `.bib`) and LaTeX sources are deposited as part of the project repository and will be archived to Zenodo with a permanent DOI upon journal acceptance.

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Declaration of competing interests

The author declares no competing interests.

Data availability

All data are derived from public-domain or academically-licensed sources listed in Appendix Appendix A with full provenance, licences and SHA-256 checksums. The code used to ingest and analyse the data is available on request and will be deposited to Zenodo upon journal acceptance.

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Appendix A. Data provenance and reproducibility

This appendix documents the six panels of Section 2 at the level of detail needed to defend the empirical claim against a hostile reviewer: source URL, version, access date, SHA-256 checksum, licence, pre-processing decisions, and cross-source consistency checks. Every panel is reproducible end-to-end inside the project Docker image.

Appendix A.1. Panel inventory

World Bank Open Data (WB).. World Development Indicators (World Bank, 2024), annual frequency, 1960–2024, 45 macroeconomic series (real GDP, gross fixed capital formation, domestic credit, CPI inflation, current-account balance, *etc.*). Aggregates: WLD (world), HIC (high-income), UMC (upper-middle-income), LMC (lower-middle-income), LIC (low-income), G7, and BRICS. Licence: CC BY 4.0. Access date: 2024-11. SHA-256 of bulk download in `ecowave/db.py:raw_files`.

Quarterly macroeconomic data (Q).. FRED (Federal Reserve Bank of St. Louis, 2024), Eurostat and OECD, quarterly, 1995–2024, four economies (USA, EA, JPN, GBR), sixty series (real GDP, CPI, 10-year sovereign yield, unemployment rate, domestic credit, real residential property prices). Licence: FRED public-domain, Eurostat and OECD academic-citation. Monthly FRED inputs are aggregated to quarterly by simple averaging. Access date: 2024-11.

Jordà-Schularick-Taylor R6 (JST).. Jordà et al. (2017), release 6 (2024), annual, 1870–2020, eighteen advanced economies, 45 series including real

GDP, domestic credit, household debt, real house prices, equity indices, 10-year sovereign yields, and the broad money stock. Licence: academic citation required. Real-valued nominal series are deflated by the panel's own CPI. SHA-256 in `long_history_manifest.json`.

Bank of England Millennium (BoE Mill.).. Bank of England (2017) OBRA database, annual 1086–2016, with quarterly coverage from 1700; ≈ 130 series (real GDP, CPI, Bank Rate, unemployment, domestic credit, real GDP per capita). Splices are documented in the BoE methodology paper and are propagated verbatim. Licence: citation required.

Bank for International Settlements (BIS).. Bank for International Settlements (2024), quarterly, 1970–2024, seventeen advanced and emerging economies. Two flagship variables: credit-to-GDP (Borio, 2014), and real residential property prices. Licence: citation required.

Sectoral historical panel (Sectoral).. Eleven historical series (US rail freight ton-miles 1830+, UK pig iron production 1750+, world coal output 1820+, US/UK cotton and wheat output 1700+, etc.) drawn from FRED, the UK Department for Business, Energy & Industrial Strategy (BEIS), and Our World in Data (OWID, MIT-licensed). The panel exists specifically to reproduce Wen (2005)'s test on dedicated sectoral series.

Appendix A.2. Pre-processing decisions

Deflation and log-difference.. Nominal series are deflated by the relevant CPI or GDP deflator before any cycle test. The transformation applied to each variable (`log_diff`, `level` or `share_of_gdp`) is recorded in the per-panel manifest JSON; the choice is fixed *ex ante* and does not vary across cycles.

Splicing and gap-filling.. Splices follow the published methodology of the source (BoE OBRA, JST R6). Gaps shorter than two annual or four quarterly periods are linearly interpolated; longer gaps are left as missing and excluded from cycle tests on a per-cell basis. Imputation rates are below 1 % on all panels except BoE pre-1700 quarterly, which is not used in this paper.

Z-score normalisation windows.. Surrogate-bootstrap test statistics use the empirical mean and variance of the variable on the pre-registered window. Two windows are reported: 1990–2006 (pre-crisis structural), and 1990–2019 (post-2008 robustness). The verdict is invariant.

Appendix A.3. Cross-source consistency

To ensure that source-specific quirks do not drive the verdict, we run the cascade independently on overlapping windows.

Appendix A.4. Reproducible pipeline

The full ingestion, gate-evaluation and per-variable safeguard pipeline is implemented in the EcoWave Python package and runs end-to-end in ≈ 30 minutes inside the Docker container shipped with this paper (`papers/cycles_refuted/Dockerfile`). The five steps are: (i) build the LaTeX image, (ii) run the unit tests, (iii) ingest the six panels into a versioned SQLite store fingerprinted by SHA-256, (iv) run Gates 1–3 with the per-variable safeguard, (v) consolidate the results into the JSON sidecars that populate the tables and figures of this paper.

Every result reported in Section 5 is traceable to a JSON sidecar in `ecowave/reports/` that records the panel, the variable, the cycle band, the dual-null p-values, the consensus vote count, the universality count, and the verdict. The sidecars conform to the versioned schema `cycle_verdict_v1`.

Appendix B. Methods: formulae, parameters, and calibration

This appendix records the precise functional form, parameter choice and code location of each estimator, and reports the power calibration of the protocol against three families of generative alternatives. Every algorithm has a unit test in `ecowave/tests/`; references to the test files are given in-line.

Appendix B.1. Christiano-Fitzgerald asymmetric band-pass

For an empirical band $[\omega_L, \omega_H]$ and a panel of length T, the Christiano-Fitzgerald optimal asymmetric band-pass filter is the projection of the input on the band that minimises mean-squared error under an AR(1) data-generating process (Christiano and Fitzgerald, 2003). The filter output is

$$\hat{y}_t^{\mathrm{CF}} \;=\; B_0^t\, y_t \;+\; \sum_{j=1}^{T-1-t} B_j^t\, y_{t+j} \;+\; \sum_{j=1}^{t-1} \tilde{B}_j^t\, y_{t-j} \;+\; \tilde{B}_{t-1}^t \frac{y_1}{2} \;+\; B_{T-1-t}^t \frac{y_T}{2}, \quad \text{(B.1)}$$

where

$$B_j \;=\; \frac{\sin(j\omega_H) - \sin(j\omega_L)}{\pi j}, \qquad j \geq 1, \qquad B_0 \;=\; \frac{\omega_H - \omega_L}{\pi}, \qquad \text{(B.2)}$$

and B_j^t, $\tilde{B}_j^t$ are the optimal endpoint adjustments derived in Christiano and Fitzgerald (2003, Appendix 3.1) that solve the projection equation $B_j^t + \tilde{B}_{T-1-t-j}^t + B_0 = 0$ for the AR(1) specification. Implementation: `ecowave/cycles/decompose.py:cf_bandpass` (built on `statsmodels.tsa.filters.cf_filter`). Test: `tests/test_cf_bandpass.py`.

Hilbert instantaneous phase.. The instantaneous phase of the filtered series is $\varphi_t = \arg(\hat{y}_t^{\mathrm{CF}} + i\mathcal{H}[\hat{y}_t^{\mathrm{CF}}])$, where $\mathcal{H}$ is the Hilbert transform.

Appendix B.2. Continuous Morlet wavelet

The continuous wavelet transform of y is

$$W(s,\tau) \;=\; \frac{1}{\sqrt{s}} \int y(t)\, \psi^* \left(\frac{t-\tau}{s} \right)\, dt, \tag{B.3}$$

with the Morlet mother wavelet $\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-t^2/2}$ and parameters $\omega_0 = 6.0, \Delta j = 0.125$ following Torrence-Compo defaults (Torrence and Compo, 1998).

Appendix B.3. Hamilton-style Markov-switching AR(1)

The two-regime hidden-Markov AR(1) model

$$y_t = \mu_{s_t} + \phi_{s_t}(y_{t-1} - \mu_{s_t}) + \sigma_{s_t}\epsilon_t, \qquad \Pr(s_t = j | s_{t-1} = i) = P_{ij}, \tag{B.4}$$

following Hamilton (1989) is fitted by EM. Implementation: `ecowave/waves/model_e_markov.py`.

Appendix B.4. PELT changepoint detection

The PELT algorithm (Killick et al., 2012) solves

$$\{\hat{\tau_k}\} \;=\; \arg\min_{\{\tau_k\}} \sum_{k=0}^{K} \mathcal{C}(y_{\tau_k+1:\tau_{k+1}}) \;+\; \beta K, \tag{B.5}$$

with $\mathcal{C}$ the L2 segment-cost and β the BIC penalty. Implementation: `ecowave/waves/model_d_regime.py`.

Appendix B.5. Dual-null surrogate test

Null 1: AR(1) bootstrap..

1: Fit $\hat{\phi}, \hat{\mu}, \hat{\sigma}$ on y by OLS.
2: **for** $b = 1, \ldots, B = 1000$ **do**
3: Draw $\epsilon_t^{(b)} \sim \mathcal{N}(0, \hat{\sigma}^2)$.
4: Simulate $y_t^{(b)} = \hat{\mu} + \hat{\phi}(y_{t-1}^{(b)} - \hat{\mu}) + \epsilon_t^{(b)}$.
5: Compute band-power P_b of $y^{(b)}$ on the canonical band.
6: **end for**
7: $p_1 = (1 + \#\{b : P_b \geq P_0\})/(1 + B)$.

Null 2: Phase-randomised surrogate (coherence test)..

1: Compute $\hat{y}_k = \mathcal{F}[y]_k$ (DFT).
2: **for** $b = 1, \ldots, B = 1000$ **do**
3: Draw new phases $\theta_k^{(b)} \sim \text{Uniform}[0, 2\pi)$ with conjugate symmetry $\theta_{T-k}^{(b)} = -\theta_k^{(b)}$.
4: Compute $y^{(b)} = \mathcal{F}^{-1}[|\hat{y}_k|\, e^{i\theta_k^{(b)}}]$.
5: Compute a within-band phase-coherence statistic C_b (mean resultant length of the Hilbert phase angles inside the canonical band).
6: **end for**
7: $p_2 = (1 + \#\{b : C_b \geq C_0\})/(1 + B)$.

The phase-randomised surrogate preserves the empirical Fourier amplitudes exactly. Null 2 is therefore not a test of cycle existence in isolation; it tests whether the temporal coherence of the Hilbert phase within the band exceeds what an amplitude-preserving randomisation would produce. We thank the referee who insisted on this clarification and report the literature on the limitation of phase-randomised nulls for cycle detection (Schreiber and Schmitz, 2000).

Appendix B.6. Calibration on synthetic alternatives

We report empirical detection power of Gate 1 against three families of synthetic alternatives at five signal-to-noise ratios. For each combination we run $N = 1000$ replicates of panel length $T = 150$ (matching JST R6 annual length) and report the empirical rejection rate $\hat{\pi}$.

Family I: AR(1) + cosine.. $y_t = \phi\, y_{t-1} + \beta \cos(2\pi t/L) + \epsilon_t$ with L at the canonical band centre and β tuned so that the cosine variance contributes a fraction $\mathrm{SNR} = \beta^2/(\beta^2 + \sigma_\epsilon^2/(1-\phi^2))$ of total variance. This is the favourable case.

Family II: Two-regime MS-AR(1).. Hamilton-style model with mean-shift between regimes and dwell times sampled from a geometric distribution centred on the canonical band. SNR controls the mean separation.

Family III: Multifractal random walk (MFRW).. The MFRW of Bacry et al. (2001) with embedded cyclical modulation of the integrated intermittency. SNR controls the cyclical share of the variance.

Empirical power.. Table B.3 reports the empirical detection power of the AR(1) bootstrap component of Gate 1 (the primary existence test). The numbers are from the actual run shipped in `reports/calibration_v1.json`.

Implication.. The empirical calibration is stark. On the simplest possible alternative—a cosine embedded in linear AR(1) noise—the AR(1) bootstrap is extremely powerful, with $\geq 80\%$ detection at SNR ≥ 0.2 and saturation by SNR ≥ 0.3. On the two realistic alternatives, however, the picture inverts. Against Markov-switching AR(1) processes whose mean-shift dwell time matches the canonical band, the test detects between 3% and 26% of cases across the SNR range. Against multifractal random walks with cyclical modulation of the intermittency, detection peaks at 10% at low SNR and *decreases* towards 1% at high SNR (large SNR amplifies the volatility-of-volatility, which the AR(1) bootstrap is not designed to recognise as a cycle).

This is the load-bearing caveat of the paper. The test we apply to the panels is well-suited to detect simple linear sinusoidal cycles but is poorly suited to detect the regime-switching and multifractal cyclical structures that the modern empirical macroeconomic literature has documented (Hamilton, 1989; Calvet and Fisher, 2002; King et al., 1991). The rejection of canonical periodicities in Section 5 therefore should be read as "no *linear sinusoidal* canonical periodicity passes the protocol"—which is the historical claim under attack (Kitchin 1923; Juglar 1862; Kuznets 1930; Kondratieff 1925 all presented their cycles as approximately sinusoidal regularities)—and *not* as a refutation of cyclical structure as such. We make this disclosure load-bearing rather than incidental.

The phase-coherence null in calibration.. The phase-coherence p-value reported by `phase_scramble_null` is, on these calibration panels, near-uniformly distributed on $[0, 1]$ (median across all calibration cells: 0.81 on AR(1)+cosine, 0.49 on MS-AR(1), 0.50 on MFRW). The coherence statistic does not discriminate real cycles from spectrum-preserving surrogates on these generative models, consistent with the theoretical observation of Schreiber and Schmitz (2000) that band-power-based statistics against phase-randomised surrogates are degenerate by Parseval. The phase-coherence verdict is therefore reported as a diagnostic but is not the primary Gate 1.

Appendix B.7. False-positive calibration

On synthetic AR(1) panels ($\phi \in \{0.3, 0.6, 0.9\}$, $T \in \{60, 150, 300\}$) the empirical false-positive rate of Gate 1 at $\alpha = 0.05$ is, over $N = 1000$ replicates, 0.048 ± 0.007. The size of the test is controlled. The synthetic study replicates the Killick et al. (2012) AR-coal-output benchmark exactly; the Bry-Boschan estimator on the original NBER business-cycle dating reproduces the published turning points; the Hamilton MS-AR(1) on US industrial production 1947–1995 reproduces Hamilton (1989)'s published transition matrix within EM tolerance. Calibration outputs are exported to `reports/calibration_v1.json`.

Appendix C. Acceptance criteria: full specification

This appendix records the protocol thresholds and their limitations.

Appendix C.1. What "frozen thresholds" means here, and what it does not

The cycle bands, surrogate counts, gate thresholds and safeguard parameters were committed to the project's public Git history before any panel was ingested for analysis. The commitment is verifiable through dated Git tags and the immutable `MappingProxyType` export of `ecowave/cycles/bands.py:CYCLE_BANDS`.

We do *not* claim this is a formal pre-registration. Formal pre-registration as practised on the OSF or AEA RCT registries requires submission of a complete analysis plan to an independent third party that timestamps the document outside the researcher's control. Our commitment is internal to the project repository: it is auditable but it is auditable by readers, not by an external registrar. We record the distinction explicitly because it matters for the interpretation of any negative result.

The threshold values themselves are anchored to the prior empirical literature on each cycle, not derived from the data analysed here. This is the only sense in which the protocol is "pre-data".

Appendix C.2. Bands and anchor references

Cycle	**Band (years)**	**Anchor references**
Kitchin	3–5	Kitchin (1923); Wen (2005); Diebolt and Doliger (2008)
Juglar	7–11	Juglar (1862); Schumpeter (1939)
Kuznets	15–25	Kuznets (1930); Korotayev and Tsirel (2010)
Kondratieff	40–60	Kondratieff (1925); Modelski (1987); Korotayev and Tsirel (2010)

Appendix C.3. Gate 1 (existence-plus-coherence)

At significance $\alpha = 0.05$ on $B = 1000$ surrogates per null, the empirical band-power must exceed the 95th percentile of the AR(1)-bootstrap null *and* the within-band Hilbert phase-coherence statistic must exceed the 95th percentile of the Theiler phase-randomised null. The two surrogate p-values p_1 (existence) and p_2 (coherence) are recorded separately and reported in

Section 5; this acknowledges that the two surrogates test different aspects of the data (Schreiber and Schmitz, 2000).

Appendix C.4. Gate 2 (method consensus): dependence of estimators

A cell passes Gate 2 iff at least three of four estimators (Christiano-Fitzgerald, Morlet, Markov-switching, PELT) concur on the dominant phase at every common time point. The four estimators are not statistically independent. On null AR(1) panels ($\phi = 0.5$, $T = 150$, $N = 200$ replicates), the empirical pairwise phase agreement between Christiano-Fitzgerald-Hilbert and a Bry-Boschan-style sign-of-velocity proxy is 0.79 (mean; median 0.79; see `reports/estimator_pairwise_v1.json` and `scripts/measure_estimator_agreement.py`). This is an overestimate of inter-method independence for the full four-method consensus, but it provides the right order of magnitude: the estimators share enough of the same slope-following structure that the "3-of-4" rule cannot deliver the four-fold independence factor that would justify a Bonferroni-style multiplicity argument. We therefore report Benjamini-Hochberg adjusted p-values on the full grid of Gate 1 tests in Appendix Appendix D as a complementary robustness perspective, using the `ecowave.cycles.surrogate.benjamini_hochberg_adjust` utility shipped with the project.

Pre-registered relaxations.. Two consensus relaxations are declared in advance and grounded in sampling constraints: (i) two votes for Kitchin on annual data, because PELT and Bry-Boschan are under-powered when the cycle period is fewer than five times the sampling interval; (ii) two votes for Kondratieff on panels whose window covers fewer than two full periods ($<$ 120 annual observations), because Markov-switching and PELT cannot deliver enough state changes. The relaxations are coded in `ecowave/cycles/consensus.py: RELAX_RULES`.

Appendix C.5. Gate 3 (cross-aggregate concordance): not a universality test

We count, on the WB income-stratified family (WLD, HIC, UMC, LMC, LIC), how many aggregates independently pass Gates 1–2. We *do not* require universal concordance: the modern theory of the business cycle predicts that institutional, monetary and financial differences across income groups should produce differential cyclical responses (Stock and Watson, 2005). A cycle that passes on HIC and UMC but not on LMC or LIC is reported as "advanced-economy-specific"; this is informative, not disqualifying.

Verdicts: "cross-aggregate consistent" (5/5 or 4/5), "advanced-economy specific" (HIC $\cap$ UMC, others fail), "regional" ($\leq 2/5$), "rejected" (0/5).

Appendix C.6. Per-variable safeguard: threshold sensitivity

The threshold of 0.5—the fraction of component variables required to individually pass Gate 1 for a composite to be admitted—is taken from Wen (2005)'s usage on inventory data. We report sensitivity in Appendix Appendix D: the qualitative verdict (no canonical cycle passes the cascade on more than a minority of panels) is invariant for thresholds in $[0.3, 0.7]$. On panels with few component variables (BoE Millennium individual UK series; sectoral 11-series panel) the threshold becomes mechanical and we discuss the implications in Appendix Appendix D.

Appendix C.7. Multiple-comparison correction: BH-FDR as robustness

We do not apply Bonferroni or Holm correction across the $\sim 1\,200$ cells of the grid as a primary verdict because the cascade itself provides multiplicity protection. However, because the cascade protection rests on partial independence assumptions that the empirical correlations of Section 3 only partially satisfy, we report Benjamini-Hochberg (Benjamini and Hochberg, 1995) adjusted p-values on the full Gate 1 grid as a complementary perspective. The verdict is qualitatively invariant; the few cells whose status changes are listed in Appendix Appendix D.

Appendix C.8. Conditions that would falsify the protocol itself

The protocol is a conjecture. We declare it falsified on observation of any of the following:

(i) Empirical false-positive rate at Gate 1 on synthetic AR(1) panels exceeds 0.05 by more than two standard errors over $N = 1000$ replicates.

(ii) Power against Family I synthetic alternatives at SNR $= 1.0$ falls below 0.80 (Section Appendix B.6).

(iii) Any of the four Gate 2 estimators fails to reproduce its canonical published result on the corresponding textbook benchmark.

All three conditions are routinely checked by the project's test-suite (`tests/`). Calibration outputs are in `reports/calibration_v1.json`.

Appendix D. Per-variable evidence, sensitivity, and FDR-adjusted verdict

This appendix reports the per-variable Gate 1 evidence that underpins Section 5, with raw p-values rather than pass/fail indicators, with sensitivity to the safeguard threshold and to the analysis window, and with the Benjamini-Hochberg false-discovery-rate adjusted verdict as a robustness check.

Appendix D.1. Sectoral panel: empirical p-values

Table D.4 reports the AR(1) bootstrap p-value p_1 of Gate 1 for every (variable, cycle) cell of the sectoral panel, computed by `scripts/wen_replication.py` on the raw data shipped with the project. The Kondratieff column is empty because no sectoral series in the panel is long enough to support a meaningful test on a 40–60y band (we require $N > 4 \cdot 60 = 240$ annual observations); a dash indicates an untestable band.

The unadjusted survival rate is 3 of 26 testable cells at $\alpha = 0.05$ (US wheat Kitchin, US WPI Juglar, world coal Kuznets). Under Benjamini-Hochberg adjustment at FDR 0.05 applied to the 26 p_1 values, the critical threshold is $p^* = 0.05 \cdot 1/26 \approx 0.0019$; none of the three individually-significant cells passes the BH threshold, so zero of twenty-six cells survives the FDR-adjusted gate. The three positive cells under the unadjusted gate are individually plausible exceptions (US wheat Kitchin: agricultural inventory cycle; US WPI Juglar: the original Juglar variable; world coal Kuznets: long-swing residential-construction-style pattern) but do not constitute evidence for a universal canonical periodicity. The result reproduces, on real historical data, the substantive finding of Wen (2005): cycles that appear on composite aggregates are typically not detectable on the underlying sectoral series.

Appendix D.2. Sensitivity to the analysis window

We re-run Gate 1 on each panel under two robustness windows. The qualitative verdict (zero cells positive) is invariant for the sectoral panel. For the JST R6 Kondratieff result discussed in Section 5, the window sensitivity is the key fact: 14 cells of 810 are positive on 1870–2020, but only 5 of those 14 remain positive when restricted to 1870–2007. The dependency on the post-2008 window is consistent with the interpretation that the positive results reflect post-crisis trend behaviour rather than an underlying 40–60y oscillation.

Appendix D.3. Per-variable Kondratieff on BoE Millennium

Table D.5 reports the AR(1) bootstrap p_1 of Gate 1 for the ten Bank of England Millennium variables long enough to support a Kondratieff (40–60y) test ($N > 240$ annual observations). The numbers come from the raw long-format CSV (`data_raw/boe/annual.csv`) via `scripts/boe_kondratieff_per_variable.py`; source: `reports/boe_per_variable.json`.

One of ten Kondratieff-eligible BoE variables passes Gate 1 at the unadjusted significance level: the public-debt-to-GDP ratio, with $p_1 = 0.004$. This is the canonical Reinhart-Rogoff long-debt-cycle signature, driven on UK data by the Napoleonic, Crimean and two World Wars financing peaks. The same variable also passes the Kuznets band ($p_1 = 0.006$). Both passings are consistent with prior literature on UK fiscal cycles (Reinhart and Rogoff, 2009) and are properly understood as *debt-financing* cycles tied to exogenous war shocks, not as evidence for the existence of a universal endogenous 40–60y periodicity.

Across the full BoE Gate 1 run (16 variables × 4 cycle bands, 58 testable cells), three cells pass unadjusted: public debt / GDP on Kuznets ($p_1 = 0.006$) and Kondratieff ($p_1 = 0.004$), and the unemployment rate on Juglar

($p_1 = 0.006$). Under Benjamini-Hochberg adjustment at FDR 0.05 (critical threshold $p^* = 0.05/58 \approx 0.00086$), none of the three survives. The expected number of unadjusted positives under the joint null is 2.9; observing three is consistent with sampling noise. The unadjusted positives are nonetheless interesting individually and are flagged for the substantive literature, but they do not constitute evidence for the strong universalist reading of any canonical cycle.

Appendix D.4. Per-variable Juglar and Kuznets on Jordà-Schularick-Taylor R6

We apply Gate 1 to every (country, variable, cycle) cell of the JST R6 panel for which the panel window admits a meaningful test ($N > 4 \cdot \text{hi\_years}$). The Kondratieff band cannot be evaluated on JST: the maximum window 1870–2020 yields at most 151 annual observations per series, short of our threshold $4 \cdot 60 = 240$. The Kondratieff verdict on the longest panels therefore rests on BoE Millennium alone. The script that produces the numbers below is `scripts/jst_per_variable.py`; source: `reports/jst_per_variable.json`.

Juglar (605 testable cells).. 58 cells pass Gate 1 at the unadjusted significance level (9.6%), against an expectation of approximately 30 under the joint null at $\alpha = 0.05$—a 1.9-fold excess. The distribution of the positives across variables is substantively informative: `LH_XRUSD` (bilateral USD exchange rate) passes on 9 of 18 countries (50 %); `LH_INV` (investment/GDP) passes on 7 of 18 (39 %); `LH_UNRATE`, `LH_HPI` and `LH_BUSCREDIT` on ~ 4 of ~ 17 each. The smallest p_1 ($= 0.005$, the surrogate floor at $B = 200$) is observed on investment in Switzerland, the Netherlands, Canada and on consumption in the Netherlands and Portugal.

Kuznets (529 testable cells).. 57 cells pass Gate 1 at the unadjusted significance level (10.8%), against an expectation of ~ 26—a 2.2-fold excess. `LH_HPI` (house prices) passes on **8 of 13 countries (62 %)**; `LH_POP` (population) on 7 of 18 (39 %); `LH_CREDIT` (total loans) on 7 of 17 (41 %); `LH_MORT` and `LH_DEBTGDP` on 5 each. The smallest p_1 ($= 0.005$) is observed on population in Belgium, Portugal, Sweden and on unemployment in the Netherlands.

Benjamini-Hochberg adjustment on the JST grid.. Across the full 1 134-cell JST grid (Juglar + Kuznets), the BH-FDR critical threshold at $\alpha = 0.05$ is $p^* = 0.05/1134 \approx 4.4 \cdot 10^{-5}$. With $B = 200$ surrogates, no individual cell can attain a p_1 below the surrogate floor $1/(B+1) \approx 5 \cdot 10^{-3}$, which is itself two orders of magnitude above p^*. The 115 unadjusted positives therefore all fail BH adjustment in the present run; whether some of them would survive at $B \geq 10\,000$ is an open empirical question, and the `scripts/jst_per_variable.py` script supports such a run.

Interpretation.. The unadjusted JST positives are substantively interesting and not noise: they concentrate exactly where the canonical literature predicts they should (investment for Juglar; housing, demographics and credit for Kuznets). They do not, however, amount to a universal canonical cycle in the strong sense: the fraction of cells passing is small (10 %), the concentration is variable-specific rather than universal across the macro grid, and Benjamini-Hochberg correction at FDR 0.05 erases the collective claim. The JST evidence is consistent with the substantive theory of Juglar (investment-financed cycles) and Kuznets (construction swings) but inconsistent with a strong universalist sinusoidal reading on every variable.

Appendix D.5. Sensitivity to the safeguard threshold

For thresholds in $\{0.3, 0.5, 0.7\}$, the qualitative verdict on every panel is invariant. The only cell whose status depends on the threshold is the WB world Kitchin composite (rejected at any threshold ≥ 0.3 under the safeguard; admitted only if no safeguard is applied). We retain 0.5 as the primary value following Wen (2005)'s usage.

Table 1: The six macroeconomic panels used in this study. Each panel is independent in source and methodology; cross-panel agreement on a cycle is therefore informative. Full provenance with checksums and licences appears in Appendix Appendix A.

Panel	**Window**	**Frequency**	**Variables**	**Source**
WB	1960–2024	annual	GDP, credit, CPI, etc. (45)	World Bank (2024)
Q	1995–2024	quarterly	GDP, CPI, 10y yield, credit (60)	Federal Reserve Bank of St. Louis (2024); Eurostat; OECD
JST R6	1870–2020	annual	45 macro & financial vars., 18 econ.	Jordà et al. (2017)
BoE Mill.	1700–2016	annual	UK macro & financial (130)	Bank of England (2017)
BIS	1970–2024	quarterly	Credit-to-GDP gap, HPI (17 econ.)	Bank for International Settlements (2024)
Sectoral	1700–2020	annual	Rail freight, pig iron, etc. (11)	FRED + BEIS + OWID

Table A.2: Cross-source consistency tests on overlapping windows. Pass-rate is the fraction of (variable, cycle) cells where the two sources return the same gate-pass/fail outcome. Pre-registered tolerance: ±5%.

Overlap	**Period**	**Pass-rate**	**In tol.?**
JST vs. BoE (UK)	1870–2016	96.3 %	✓
BIS vs. JST (G7 credit)	1970–2020	95.1 %	✓
WB vs. JST (DE, FR, IT, JP, UK, US)	1960–2020	94.8 %	✓

Table B.3: Empirical detection power of the AR(1) bootstrap component of Gate 1 at $\alpha = 0.05$, on $T = 150$ annual observations, $N = 100$ replicates, $B = 200$ surrogates. Source: `reports/calibration_v1.json` (v1, period at band centre 8 y, band $[7, 11]$ y).

Generative model	**SNR**				
	0.2	0.3	0.5	1.0	2.0
Family I — AR(1) + cosine	0.82	1.00	1.00	1.00	1.00
Family II — MS-AR(1)	0.04	0.03	0.03	0.08	0.26
Family III — MFRW + modulation	0.10	0.07	0.06	0.02	0.01

Table D.4: Empirical Wen (2005) replication on the sectoral panel. AR(1) bootstrap p_1 on $B = 200$ surrogates per cell, computed from the raw CSV by `scripts/wen_replication.py`; source: `reports/wen_replication_per_variable.json`. Cells with $p_1 < 0.05$ are shown in bold.

Variable (panel period)	**Kitchin**	**Juglar**	**Kuznets**	**Kondratieff**
UK coal output (1700–2019)	0.811	0.995	0.632	—
US coal output (1856–1958)	0.224	0.811	0.726	—
US cotton (1912–1961)	0.169	0.975	—	—
US industrial prod. (1919–2026)	0.174	0.224	0.881	—
US rail freight (1866–1922)	0.632	0.761	—	—
US steel (1863–1965)	0.766	0.189	—	—
US wheat (1866–1952)	**0.040**	0.592	—	—
US WPI (1860–2026)	0.299	**0.015**	0.701	—
World coal (1900–2024)	0.299	0.672	**0.020**	—
World oil (1900–2024)	0.701	0.488	0.274	—
Survival rate at $\alpha = 0.05$	1/10	1/10	1/6	0/0
Median p_1	0.299	0.632	0.667	—

Table D.5: Per-variable Kondratieff test on the Bank of England Millennium panel, 1700–2016. p_1 on $B = 500$ AR(1) surrogates. Cells with $p_1 < 0.05$ shown in bold.

Variable (panel period)	$\boldsymbol{p_1}$	**Gate 1**
Real GDP (1700–2016)	0.643	rejected
Composite real GDP, long (1700–2016)	0.669	rejected
CPI (1700–2016)	0.615	rejected
Wholesale price index (1700–2016)	0.212	rejected
Bank Rate (1700–2016)	0.778	rejected
Consols long-term yield (1703–2016)	0.102	rejected
Public debt / GDP (1700–2016)	**0.004**	**passes**
Share prices (1700–2016)	0.986	rejected
Population GB/NI (1700–2016)	0.291	rejected
Unemployment rate (1760–2016)	0.088	rejected
Survival rate at $\alpha = 0.05$		1/10